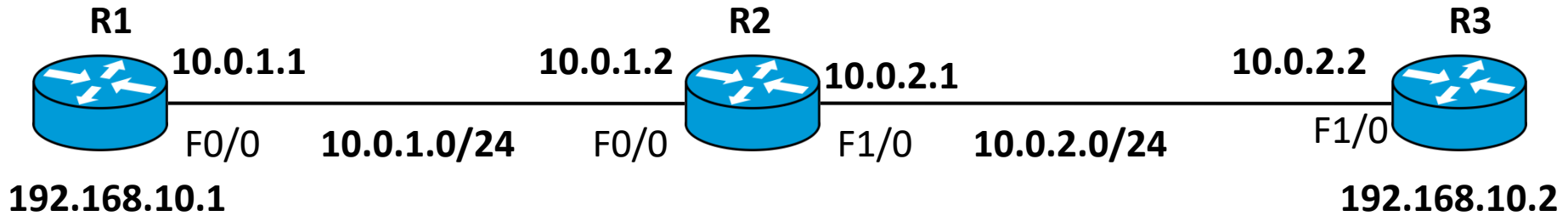


EIGRP Route Metric lab:



Explanation:

Three Routers are there and then we will calculate the Route Metric, further which is used to calculate best path.

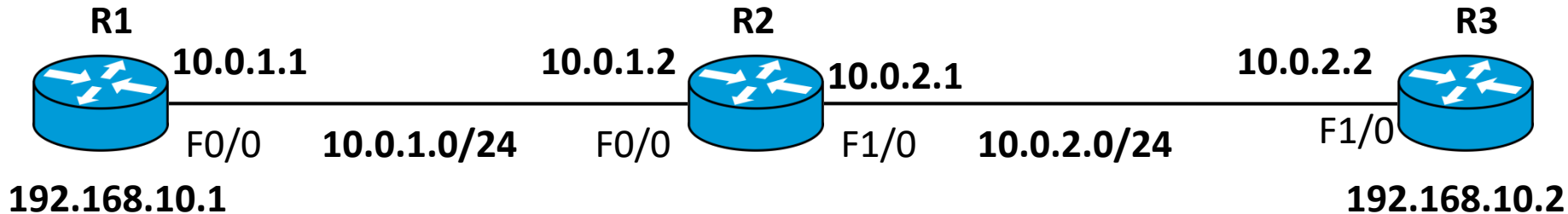
There are three steps to do that:

STEP 1: First of All Provide IP Address

STEP 2: Configure the Routing

STEP 3: Calculate Route Metric

EIGRP METRIC FORMULA:

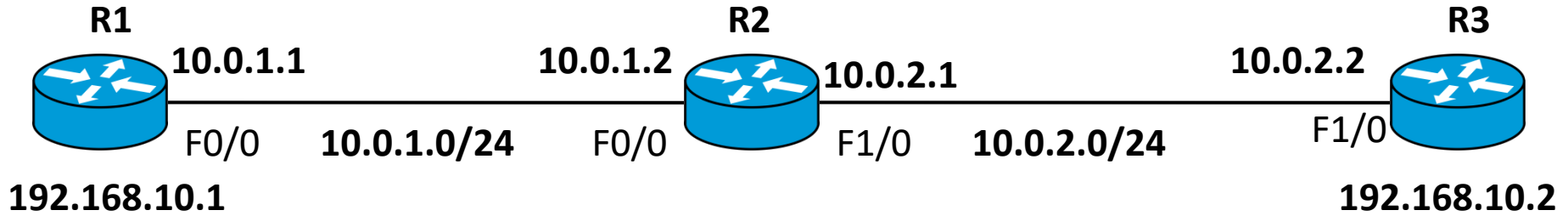


$$\text{Metric} = 256 \times \left[K1 \times \text{Bandwidth} + \frac{K2 \times \text{Bandwidth}}{256 - \text{Load}} + K3 \times \text{Delay} \right]$$

Simplified Formula, When $K1 = 1$, $K3 = 1$, $K2 = K4 = K5 = 0$

$$\text{Metric} = 256 \times (\text{Bandwidth} + \text{Delay})$$

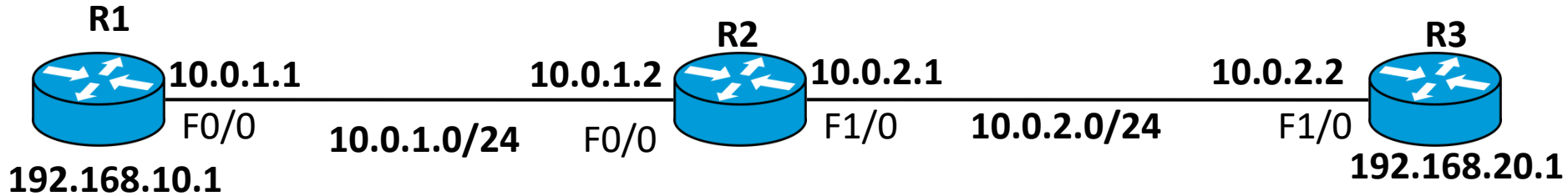
Explanation of Variables: EIGRP Route Metric



$$\text{Bandwidth} = \frac{10^7}{\text{Least Bandwidth}}$$

$$\text{Delay} = \frac{\text{Sum of All Interface Delay (in } \mu\text{s)}}{10}$$

Step 1: Check Routing Table: EIGRP Route Metric



R1#show ip route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks

C 10.0.1.0/24 is directly connected, FastEthernet0/0

L 10.0.1.1/32 is directly connected, FastEthernet0/0

D 10.0.2.0/24 [90/30720] via 10.0.1.2, 00:01:07, FastEthernet0/0

192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.10.0/24 is directly connected, Loopback1

L 192.168.10.1/32 is directly connected, Loopback1

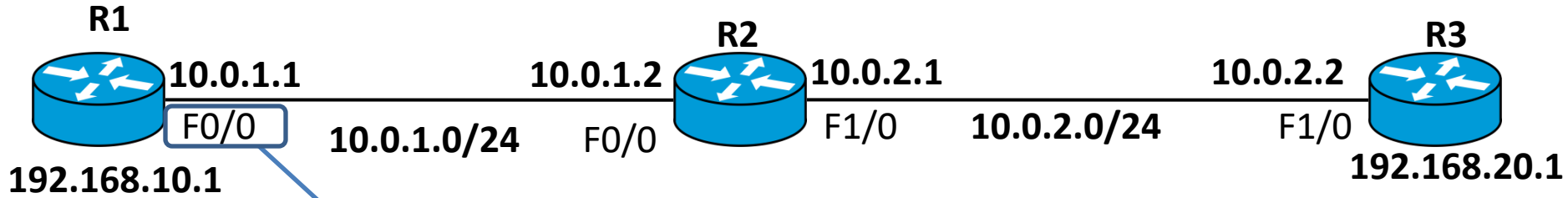
D 192.168.20.0/24 [90/158720] via 10.0.1.2, 00:00:47, FastEthernet0/0

Route Metric to Reach

R3 Loopback 192.168.20.1

Is 158720

STEP 2: Check Bandwidth and Delay on R1: EIGRP

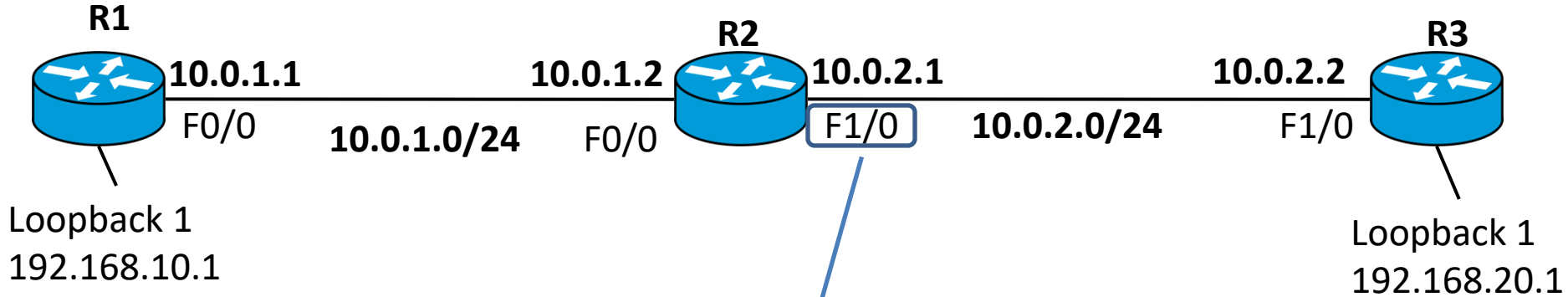


Check Bandwidth and Delay on R1 Interface F0/0

R1#show int f0/0 | Section DLY

MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255

STEP 2: Check Bandwidth and Delay on R2 EIGRP

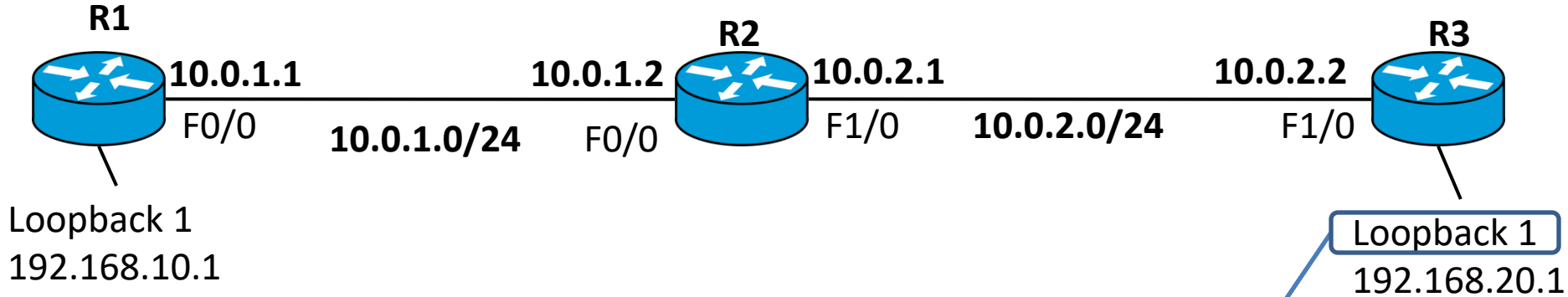


Check Bandwidth and Delay on R2 Interface F1/0

R2#show int f1/0 | Section DLY

MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255

STEP 2: Check Bandwidth and Delay on R3: EIGRP

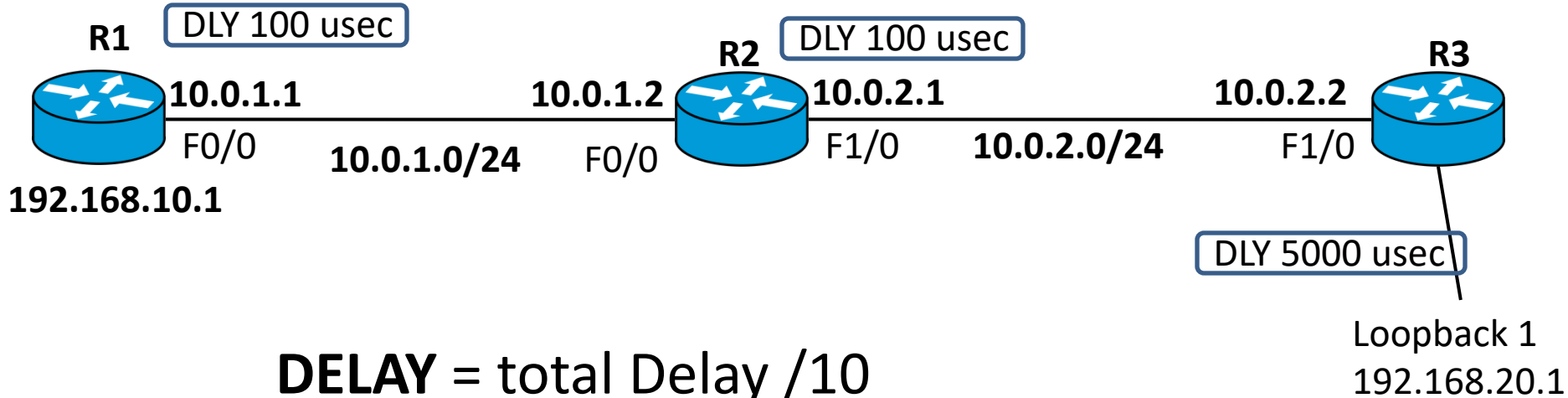


Check Bandwidth and Delay on R3 Interface Loopback 1

R3#show int loopback 1 | Section DLY

MTU 1514 bytes, BW **8000000** Kbit/sec, DLY **5000** usec,
reliability 255/255, txload 1/255, rxload 1/255

EIGRP Calculate Cumulative Delay:-

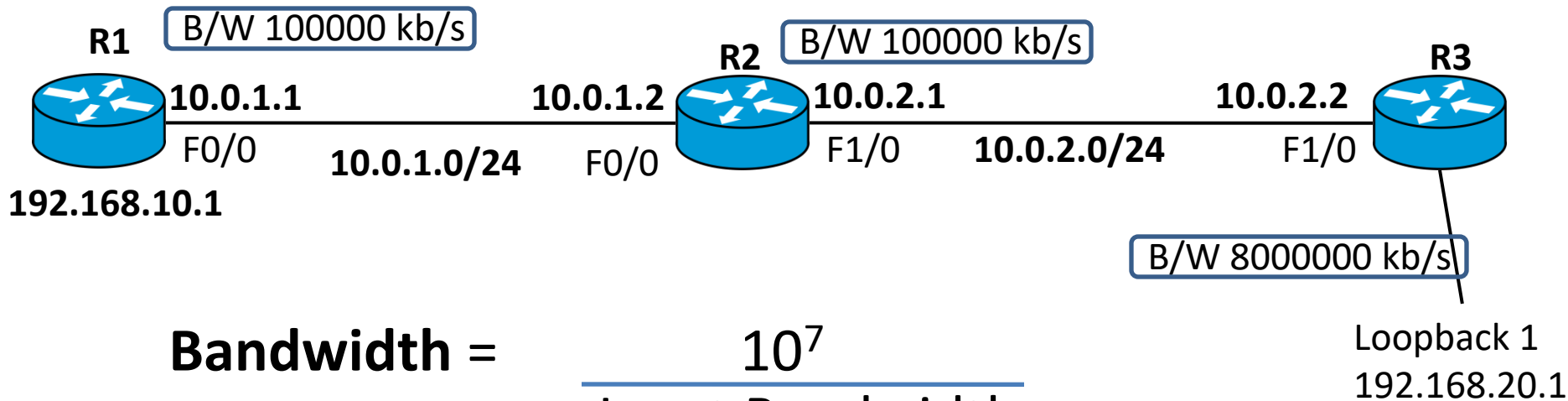


DELAY = total Delay /10

DELAY = 5000+100+100/10

DELAY = 520

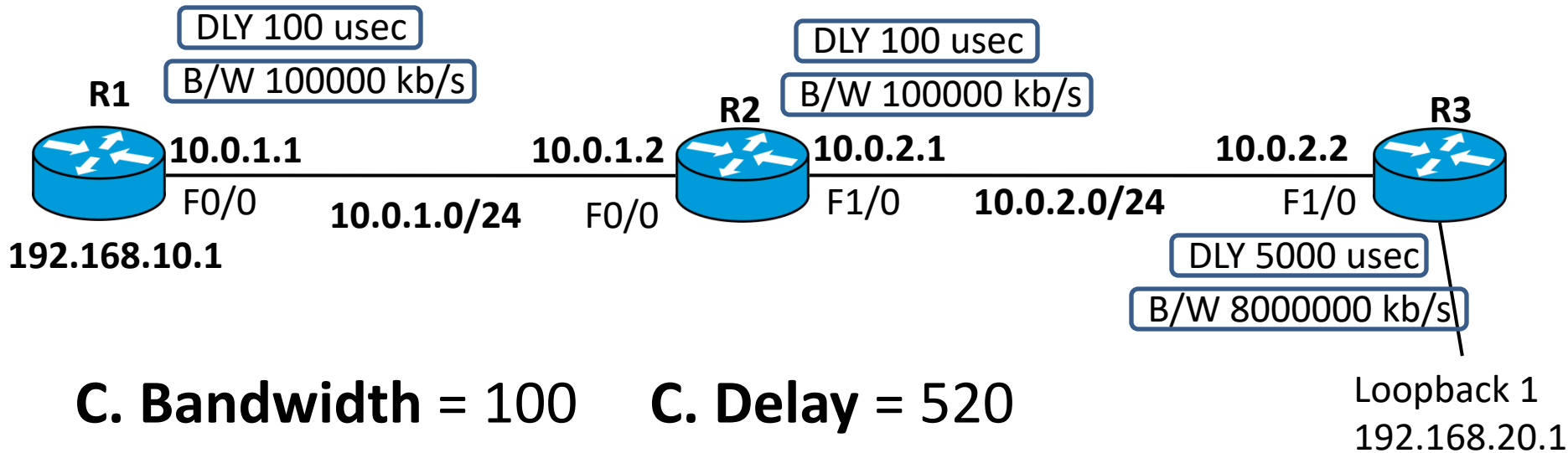
EIGRP Calculate Cumulative Bandwidth: -



$$\text{Bandwidth} = \frac{10^7}{\text{Least Bandwidth}}$$

$$\text{Bandwidth} = \frac{10^7}{100000} = 100$$

EIGRP: Calculate Route Metric: -



C. Bandwidth = 100 C. Delay = 520

Route Metric = $256 \times (520 + 100) = 158,720$

If Multiple Path are there, the path having least Route metric will be the best path